

nTZS Issuance & Redemption Protocol

Prepared by: NEDA LABS Company Limited **For:** Bank of Tanzania — Fintech Regulatory Sandbox (Ref. LD.170/515/02/1254)
Submission item: §7(c) — issuance and redemption protocol, diagrammatic token flow, and smart-contract control documentation
Status: Draft for counsel / regulator review · *fields in [brackets] to be completed* **Version:** 0.1 · [date]

1. Purpose & scope

This document describes how the nTZS tokenised stablecoin is **issued, transferred, redeemed, and used in cross-currency settlement** during the Bank of Tanzania (BoT) Fintech Regulatory Sandbox test, and the controls that keep it compliant with the approved Testing Parameters.

nTZS is a Tanzanian-Shilling-pegged digital token, backed on a strict **1:1 basis** by TZS-denominated reserves (cash and short-term government securities) held in a **ring-fenced trust/escrow account** at a BoT-licensed commercial bank. nTZS is **not legal tender**, is restricted to approved sandbox participants, and **all customer-facing settlement is in TZS**.

2. Parties & roles

Party	Role	Sits in	Permission
NEDA LABS Co. Ltd	Issuer/administrator of nTZS; controls mint & burn; operates the wallet/payment UX	Tanzania	BoT sandbox participant
[Partner Bank] (BoT-licensed)	Trustee/custodian of TZS reserves; FX principal (TZS↔USD); TZS settlement; AML/CFT	Tanzania	Banking & FX licence (letter §6)
[Offshore VASP]	Facilitates the USDC float; USD↔USDC via Circle; on-chain USDC counterparty for cross-border	[Jurisdiction]	Foreign VASP/MSB licence — outside the sandbox
[PSP(s)] (e.g. Snippe, AzamPay)	Mobile-money / bank pay-in and pay-out of TZS	Tanzania	BoT-licensed PSP

No single party performs both the TZS reserve function and the USDC function; each acts only within its own licence and jurisdiction.

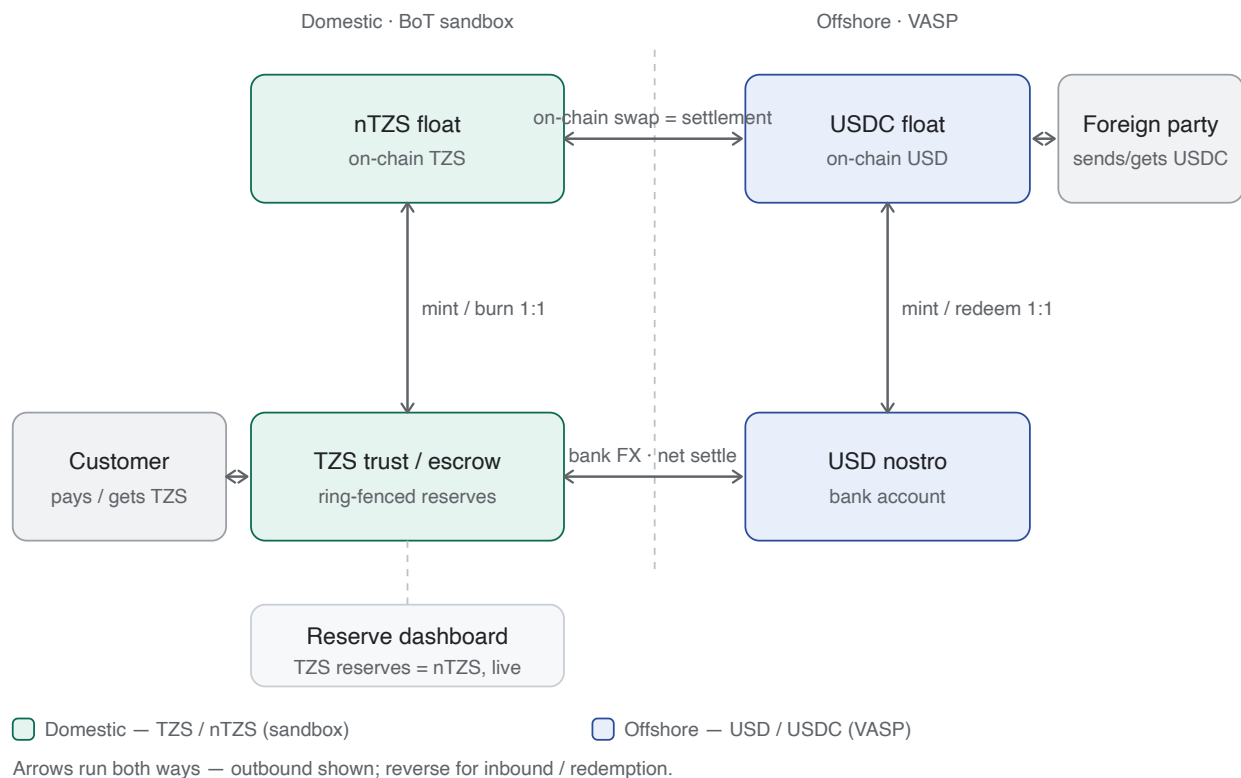
3. The two settlement clocks (overview)

The corridor runs on two decoupled "clocks":

- **Fast clock (on-chain, ~seconds):** cross-currency value moves as an instant `nTZS ↔ USDC` swap between two 1:1 reserve-backed instruments. Because each is fully backed, the swap *is* the settlement.
- **Slow clock (regulated fiat, net/batched):** the underlying reserves — TZS in the bank's trust, USD in the nostro — rebalance the floats periodically, net of inbound and outbound, through the bank's licensed FX function.

nTZS ↔ USDC corridor — two settlement clocks

On-chain swap settles in seconds; the regulated reserves net behind it.



4. Issuance (mint) — strictly after cash

1. Customer initiates a TZS pay-in via a licensed **PSP** (mobile money or bank).
2. Funds settle into the **ring-fenced trust account** at [Partner Bank]. The bank confirms receipt.
3. **Only after** confirmed receipt of the equivalent TZS in trust, NEDA's issuance contract **mints nTZS 1:1** to the customer's wallet.
4. The mint event and the reserve increment are reflected in the **real-time reserve dashboard** (§7), keeping **TZS reserves = nTZS issued** at all times.

Control: nTZS is never minted before the equivalent cash is in trust (no "fake electronic money"). The mint authority is technically gated on a confirmed-deposit reference; see §8.

5. Redemption (burn)

1. Holder requests redemption (or a TZS payout).
2. NEDA's contract **burns** the nTZS.
3. The equivalent TZS is **released from the trust account** by [Partner Bank] and paid to the beneficiary **in TZS** via PSP/bank rails.
4. The dashboard decrements issued supply against reserves, preserving the 1:1.

6. Cross-currency settlement (FX)

When value crosses currency (e.g. inbound USD/USDC → TZS payout, or TZS → USD/USDC abroad):

1. The **bank's FX desk is the principal and price-setter** for TZS↔USD — the regulated FX occurs onshore, under BoT's purview.

- The on-chain leg is an **atomic delivery-versus-delivery (DvP)** swap: NEDA delivers/burns the **nTZS** leg; [Offshore VASP] delivers/receives the **USDC** leg; the exchange executes **at the bank's rate** in a single transaction.
- Because the swap is atomic, neither party holds the other's instrument beyond the instant of settlement — **no foreign stablecoin enters Tanzania, and no unbacked nTZS leaves**.
- All USD↔USDC activity (Circle Mint / OTC) is performed **offshore** by [Offshore VASP]. The only domestic touchpoints are the bank's USD nostro (a normal correspondent flow) and the atomic on-chain settlement.
- Inbound and outbound flows **net against each other**, so the working floats stay small and no party pre-funds gross volume.

7. Reserve management & 1:1 backing

- Customer funds are held in a **single, ring-fenced trust/escrow account** at [Partner Bank], segregated from NEDA's operational funds; no rehypothecation; no yield/staking.
- A **real-time reserve dashboard** provides BoT and the parties with continuous visibility of **fiat reserves vs. tokens issued**. Target invariant at all times: $TZS \text{ reserves} \geq nTZS \text{ issued}$.
- Independent reconciliation runs continuously (reserve balance ↔ on-chain token supply), with automated alerting on any divergence beyond tolerance.
- Bank-issued reserve attestations are produced [frequency] for the record.

8. Smart-contract control points

Control	Mechanism
Mint authority	Restricted to NEDA's issuance service, gated on a confirmed trust-account deposit reference. No open minting.
Burn authority	Restricted to redemption/payout flow; burns are recorded against released TZS.
Swap (DvP)	Fixed-rule atomic exchange at the bank-provided rate; no discretionary pricing by NEDA; per-transaction and inventory limits enforced on-chain.
Pause / kill-switch	The parties (and BoT, on direction) can halt minting and/or swaps; redemptions remain available.
Limits	Per-transaction and aggregate caps; rate-staleness guard (no fill on a stale price).
Auditability	Every mint/burn/swap is on-chain and timestamped; off-chain references link to PSP and bank records.

Full contract addresses, ABIs, role assignments, and limit parameters are provided in the accompanying smart-contract control schedule [attachment].

9. Compliance mapping (to the BoT approval letter)

Step	Rule satisfied
Customer pays/receives TZS	§4 — customer-facing settlement is in TZS
Ring-fenced trust account	§6 — TZS reserve custody, single controlled account
Mint only after cash	§4 peg + §7(d) risk plan (no fake e-money)
On-chain swap among participants	§4 — not legal tender / not a parallel currency; restricted to participants
Bank as FX principal	§6 — FX conversion is the bank's function
USDC offshore only	keeps §4 scope (domestic stays TZS-only)
Reserve dashboard	§6 — real-time reserves vs. tokens
Floats earn nothing	§4 — no staking/yield without approval

10. Glossary

- **DvP (delivery-versus-delivery):** an atomic on-chain exchange where both legs settle together or not at all.
 - **Float:** a small working balance (nTZS or USDC) sized to net intra-cycle imbalance, not gross volume.
 - **Net settlement:** periodic rebalancing of fiat reserves for the net of inbound vs. outbound, rather than per-transaction wires.
 - **VASP:** Virtual Asset Service Provider, licensed for digital-asset activity in its jurisdiction.
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This protocol is read together with the nTZS Risk Management Plan (§7(d)) and the reserve-dashboard specification.